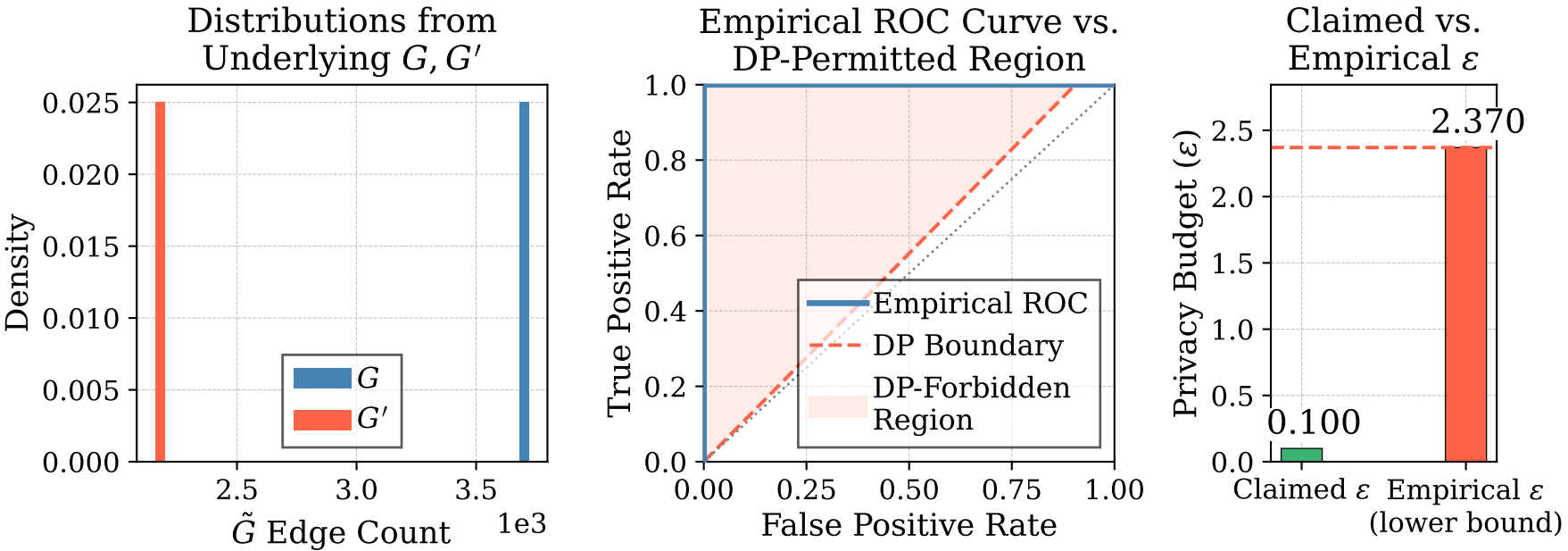
Construction-B (Undirected) ($\varepsilon = 0.1$, $\delta = 1\text{e-}05$)
Distributions from Underlying G, G'
Density
0.025
0.020
0.015
0.010
0.005
0.000
2.5
3.0
3.5
1e3
$\tilde{G}$ Edge Count
G
G'
Empirical ROC Curve vs. DP-Permitted Region
True Positive Rate
1.0
0.8
0.6
0.4
0.2
0.0
0.00
0.25
0.50
0.75
1.00
False Positive Rate
Empirical ROC
DP Boundary
DP-Forbidden Region
Claimed vs. Empirical ε
Privacy Budget (ε)
2.5
2.0
1.5
1.0
0.5
0.0
2.370
0.100
Claimed ε
Empirical ε (lower bound)